

Thursday — Physical Sciences + Python

Significant Figures · Python Setup · Variables · Operators · print() and f-strings

1 — Significant Figures

The 5 Rules

1	All non-zero digits are significant.	345 → 3 sig figs
2	Zeros between non-zero digits are significant.	3 045 → 4 sig figs
3	Leading zeros are NOT significant.	0.0045 → 2 sig figs
4	Trailing zeros AFTER a decimal ARE significant.	3.400 → 4 sig figs
5	Trailing zeros WITHOUT a decimal — ambiguous. Use scientific notation.	1 200 → could be 2, 3 or 4. Write 1.20×10^3 for 3.

Calculations

- **Adding / Subtracting:** answer has the fewest DECIMAL PLACES of any number in the problem.
- **Multiplying / Dividing:** answer has the fewest SIG FIGS of any number in the problem.

■ Memory hook:

Add/Subtract → count DECIMAL PLACES

Multiply/Divide → count SIG FIGS

Worked Examples

Example 1 — Counting sig figs: How many sig figs in 0.00460?

0.00460
Leading zeros (0.00) → NOT significant
4 → significant
6 → significant
Trailing zero after decimal (0) → significant
Answer: 3 sig figs

Example 2 — Rounding: Round 8.4752 to 3 sig figs.

8.4752 → keep 3 sig figs → look at digit 4: 8.47 52
The digit after the cut (5) is ≥ 5 → round up
Answer: 8.48

Example 3 — Addition: $12.4 + 3.75 + 0.825$

$12.4 \rightarrow 1$ decimal place
$3.75 \rightarrow 2$ decimal places
$0.825 \rightarrow 3$ decimal places
Least decimal places = 1 $\rightarrow$ round answer to 1 d.p.
$12.4 + 3.75 + 0.825 = 16.975 \rightarrow$ round to 1 d.p. = 17.0
Answer: 17.0

Example 4 — Multiplication: 4.56×1.4

$4.56 \rightarrow 3$ sig figs
$1.4 \rightarrow 2$ sig figs
Fewest sig figs = 2 $\rightarrow$ round answer to 2 sig figs
$4.56 \times 1.4 = 6.384 \rightarrow$ round to 2 sig figs = 6.4
Answer: 6.4

Significant Figures — Practice Questions

Part A — Count the sig figs

- Q1. How many sig figs in **0.00340**?
- Q2. How many sig figs in **10 500**?
- Q3. How many sig figs in **7.0040**?
- Q4. How many sig figs in **0.070**?
- Q5. How many sig figs in **6 000.0**?

Part B — Round to the given number of sig figs

- Q6. Round **0.008 476** to 2 sig figs
- Q7. Round **34 762** to 3 sig figs
- Q8. Round **1.005 7** to 3 sig figs
- Q9. Round **0.999 50** to 3 sig figs
- Q10. Round **2 874 500** to 4 sig figs

Part C — Calculations (give answer to correct sig figs / decimal places)

Q1 12.11 + 18.0 + 1.013
1.

Q1 0.0045 × 2 500
2.

Q1 1 840 ÷ 1.5
3.

Q1 22.4 – 2.35
4.

Q1 $3.4 \times 10^2 \times 2.10 \times 10^3$
5.

2 — Python: Day 1

Topic: Installation and Setup

1. Go to python.org/downloads → click the big yellow button for Python 3.x
2. Run the installer. **Tick 'Add Python to PATH'** before clicking Install.
3. Go to code.visualstudio.com → download and install VSCode.
4. Open VSCode → Extensions (Ctrl+Shift+X) → search **Python** → install Microsoft extension.
5. Open Terminal in VSCode (Ctrl+`) → type: `python --version` → should show 3.x.x
6. Create a folder called **sgl_dev** → File → Open Folder in VSCode.
7. New file → name it **day1.py** → you are ready.

■ Test it works: type `print('SGL online')` in `day1.py` → press F5.
If you see SGL online in the terminal — setup is complete.

Topic: Variables

A variable is a named container that stores a value. Python automatically detects the data type.

Type	Stores	Example	Notes
int	Whole numbers	<code>age = 19</code>	No decimal point
float	Decimal numbers	<code>mass = 9.81</code>	Has a decimal
str	Text	<code>name = 'SGL'</code>	Use quotes
bool	True / False	<code>active = True</code>	Capital T / F

Worked Example — Variables

```
age = 19 # int
```

```

mass = 9.81 # float
name = 'Kelvin' # str
active = True # bool

print(type(age)) #
print(type(mass)) #
print(type(name)) #
print(type(active)) #

```

Topic: Arithmetic Operators

Operator	Symbol	Example	Result
Addition	+	10 + 3	13
Subtraction	-	10 - 3	7
Multiplication	*	10 * 3	30
Division	/	10 / 3	3.333...
Floor division	//	10 // 3	3
Modulus (remainder)	%	10 % 3	1
Exponent (power)	**	10 ** 3	1000

■ // and % are the ones people forget.

// = how many whole times does it divide? 17 // 5 = 3

% = what is left over? 17 % 5 = 2

Worked Example — Operators

```

x = 17
y = 5

print(x + y) # 22
print(x - y) # 12
print(x * y) # 85
print(x / y) # 3.4
print(x // y) # 3 ← whole number part only
print(x % y) # 2 ← remainder
print(x ** y) # 1419857 (17 to the power of 5)

```

Topic: print() and f-strings

print() outputs to the terminal. f-strings embed variable values directly in a string.

Basic print:

```
print("Hello, world!")  
print("SGL Developments")
```

f-string — put f before the quote, use { } for variables:

```
name = "Kelvin"  
age = 19  
gpa = 3.87  
  
print(f"Name: {name}")  
print(f"Age: {age}")  
print(f"{name} is {age} years old.")  
print(f"GPA to 1 d.p.: {gpa:.1f}")
```

Format specifiers:

Specifier	Meaning	Example output
:.2f	2 decimal places	f'{3.14159:.2f}' → '3.14'
:d	Integer	f'{42:d}' → '42'
:e	Scientific notation	f'{12345.6:.2e}' → '1.23e+04'

Worked Example — f-strings

```
# Without f-string (clunky):  
speed = 340.5  
print("Speed: " + str(speed) + " m/s")  
  
# With f-string (clean):  
print(f"Speed: {speed} m/s")  
print(f"Speed rounded: {speed:.1f} m/s")  
  
# Output:  
# Speed: 340.5 m/s  
# Speed rounded: 340.5 m/s
```

Python — Practice Questions

Part A — What type is each variable?

Q1 What type is **speed = 340.5**?
6.

Q1 What type is **planet = 'Mars'**?
7.

Q1 What type is **count = 7**?
8.

Q1 What type is **active = False**?
9.

Q2 What does **type(3.0)** return?
0.

Part B — Predict the output

Q2 What is **17 // 4**?
1.

Q2 What is **17 % 4**?
2.

Q2 What is **2 ** 8**?
3.

Q2 What is **15 / 4**?
4.

Q2 What is **15 // 4**?
5.

Part C — Write the code

Write Python code for each task. Use the space on the right.

Q2 Store your name in a variable called **name** and print it.
6.

Q2 Store the speed of light (299 792 458) in **c** and print it.
7.

Q2 Calculate **2 to the power of 10** and print the result.
8.

Q2 Create **length = 12.5** and **width = 8.0**. Calculate and print the area.
9.

Q3 Use an f-string to print: **The area is 100.0 m2** using variables.
0.

ANSWER KEY

Check your work here — only look after you have attempted every question.

Significant Figures — Part A (Count sig figs)

Q1.	0.00340	3 sig figs
Leading zeros don't count. 3, 4, trailing zero after decimal = 3		
Q2.	10 500	3 sig figs
Trailing zeros without decimal are ambiguous. Count 1, 0, 5 = 3		
Q3.	7.0040	5 sig figs
7, 0 (between), 0 (between), 4, trailing zero = 5		
Q4.	0.070	2 sig figs
Leading zeros don't count. 7, trailing zero = 2		
Q5.	6 000.0	5 sig figs
Decimal present → all zeros count. 6, 0, 0, 0, 0 = 5		

Significant Figures — Part B (Rounding)

Q6.	Round 0.008 476 to 2 sf	0.0085
First 2 sig figs: 8, 4 → next digit 7 $\geq$ 5 → round up → 0.0085		
Q7.	Round 34 762 to 3 sf	34 800
Keep 3, 4, 7 → next digit 6 $\geq$ 5 → round up → 34 800		
Q8.	Round 1.005 7 to 3 sf	1.01
Keep 1, 0, 0 → next digit 5 $\geq$ 5 → round up → 1.01		
Q9.	Round 0.999 50 to 3 sf	1.00
Keep 9, 9, 9 → next digit 5 → round up → 1.00		
Q10.	Round 2 874 500 to 4 sf	2 875 000

Keep 2, 8, 7, 4 → next digit 5 → round up →
2 875 000

Significant Figures — Part C (Calculations)

Q1 1.	$12.11 + 18.0 + 1.013$	31.1
Decimal places: 2, 1, 3 → least = 1 → round to 1 d.p. → 31.123 → 31.1		
Q1 2.	$0.0045 \times 2\,500$	11
Sig figs: 2, 2 → answer to 2 sf → 11.25 → 11		
Q1 3.	$1\,840 \div 1.5$	1\,200
Sig figs: 3, 2 → answer to 2 sf → 1226.7 → 1\,200		
Q1 4.	$22.4 - 2.35$	20.1
Decimal places: 1, 2 → least = 1 → 20.05 → 20.1 (note: 20.0 also acceptable)		
Q1 5.	$3.4 \times 10^2 \times 2.10 \times 10^3$	7.1×10^5
Sig figs: 2, 3 → answer to 2 sf → 714\,000 → 7.1×10^5		

Python — Part A (Variable types)

Q1 6.	speed = 340.5	float
Has a decimal point → float		
Q1 7.	planet = 'Mars'	str
Text in quotes → str		
Q1 8.	count = 7	int
Whole number, no decimal → int		
Q1 9.	active = False	bool
True or False value → bool		

Q2 0.	<code>type(3.0)</code>	float
3.0 has a decimal point even though value is whole → float		

Python — Part B (Predict output)

Q2 1.	<code>17 // 4</code>	4
Floor division: $4 \times 4 = 16$, $17 - 16 = 1$ left over, whole result = 4		
Q2 2.	<code>17 % 4</code>	1
Modulus: $17 = (4 \times 4) + 1 \rightarrow$ remainder = 1		
Q2 3.	<code>2 ** 8</code>	256
$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$		
Q2 4.	<code>15 / 4</code>	3.75
Regular division always gives a float result		
Q2 5.	<code>15 // 4</code>	3
Floor division: $3 \times 4 = 12$, remainder 3, whole result = 3		

Python — Part C (Write the code)

Q2 6.	Store name and print	<code>name = "Kelvin" print(name)</code>
Q2 7.	Store speed of light	<code>c = 299_792_458 print(c)</code>
Underscores allowed for readability		
Q2 8.	2 to the power of 10	<code>print(2 ** 10)</code>
Output: 1024		
Q2 9.	Calculate area	<code>length = 12.5 width = 8.0 area = length * width print(area)</code>
Output: 100.0		
Q3 0.	f-string area print	<code>print(f"The area is {area} m2")</code>

Output: The area is 100.0 m²